Project Proposal

Data Visualization

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Basic Information

Project Title - Dominance Redefined: Visualizing Max Verstappen’s 2023 Formula 1 Season

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GitHub Repo link - https://github.com/Varun1421/Verstappen

GitPages link - https://varun1421.github.io/Verstappen/

Background & Motivation

For this project, I intend to analyze Max Verstappen’s 2023 Formula 1 season through data visualization. I selected this topic due to my personal interest in Formula 1 and Verstappen is my favorite driver. Beyond personal interest, the 2023 season presents a compelling case for visualization as it provides a clear narrative of dominance, consistency, and competitive separation throughout a single championship year.

This topic is particularly appealing from a visualization perspective because Formula 1 data inherently supports various forms of analysis: race-by-race progression, comparisons with other drivers, cumulative point accumulation, finishing consistency, and qualifying versus race outcomes. Verstappen’s 2023 season is especially suitable because it enables me to examine not only his frequency of victories but also the evolution of his performance over time and its comparison with the rest of the grid.

My motivation is to transform a sports season that fans recall as “dominant” into a measurable and explorative data set. Rather than merely asserting Verstappen’s dominance, I aim to demonstrate how that dominance manifested in the data: through finishing positions, cumulative points, consistency across races, and the widening disparity between him and his competitors. Additionally, I seek to provide users with interactive capabilities to comprehend the location of his advantage’s strength and whether that dominance was consistent or concentrated in specific calendar periods.

Project Objectives

Objective 1: Understanding Verstappen’s Dominance Across the 2023 Season: Analyze how Max Verstappen’s dominance developed by examining race-by-race point accumulation and identifying when he separated from the rest of the field.

Objective 2: Evaluating Driver vs. Car Influence: Compare Verstappen and Sergio Pérez across key performance metrics to determine whether Verstappen’s success was primarily driven by driver performance rather than car advantage.

Objective 3: Contextualizing the 2023 Season Within Formula 1 History: Compare Verstappen’s 2023 season with other dominant seasons in Formula 1 history to evaluate how it ranks across metrics such as win rate, podium frequency, and consistency.

Objective 4: Placing Verstappen Among Cross-Sport Greats: Extend the analysis beyond Formula 1 by comparing Verstappen’s 2023 season to peak performances from elite athletes in other sports to present dominance in a broader context.

Data

Primary Formula 1 Dataset:

The core dataset will consist of publicly available Formula 1 race and standings data, primarily sourced from Kaggle (2023 F1 Season (JDAustralia)) and supplementary Formula 1 statistical databases. This dataset includes detailed race-by-race results for the 2023 Formula 1 season, including finishing positions, points, qualifying results, podiums, and fastest laps.

Historical Formula 1 Benchmark Data:

To support historical comparisons, additional data from dominant Formula 1 seasons —including Michael Schumacher (2004), Sebastian Vettel (2013), and Lewis Hamilton (2019)—will be collected from publicly available Formula 1 statistics repositories.

Cross-Sport Comparative Data:

To support broader dominance comparisons, performance data for selected elite athletes in other sports will be compiled manually from publicly available sources. Since no unified dataset exists across sports, relevant metrics will be constructed from scratch to enable meaningful comparison across different competitive contexts.

How the Dataset Will Be Used:

The dataset will be used to:

- Track race-by-race progression of the 2023 Formula 1 season
- Compare Verstappen's performance directly against Sergio Pérez

- Benchmark Verstappen’s season against historically dominant Formula 1 seasons
- Normalize and compare dominance metrics across athletes from different sports

These datasets collectively provide the breadth necessary to analyze Verstappen’s 2023 season from multiple perspectives: within-season dominance, teammate comparison, historical Formula 1 benchmarking, and broader cross-sport contextualization. Their combination enables the project to move beyond simple seasonal analysis and frame Verstappen’s performance within a larger narrative of athletic dominance.

Data Processing

Because the project incorporates data from multiple sources and sports contexts, moderate preprocessing and transformation will be required before visualization.

Data Cleaning:

The Formula 1 datasets are already relatively well structured, but preprocessing will still be required to ensure consistency across sources:

- Verify that all race and standings data correspond to the intended seasons
- Standardize driver names, race identifiers, and season labels across datasets
- Sort races chronologically to preserve proper temporal ordering
- Remove irrelevant or redundant columns not used in analysis

Historical Formula 1 benchmark data will undergo similar cleaning to ensure metric consistency across seasons.

Derived Quantities

Several derived metrics will be computed to support analysis, including:

- Cumulative points progression by race
- Win rate and podium percentage
- Average finishing position
- Qualifying-to-race performance deltas
- Driver-to-driver performance gaps between Verstappen and Pérez

These derived values will support both within-season and historical comparisons.

Cross-Sport Metric Construction and Normalization

Because no unified dataset exists for comparing dominance across different sports, cross-sport athlete data will be manually compiled from public statistical sources and transformed into comparable performance metrics.

To enable meaningful comparison between athletes competing in fundamentally different sports, performance measures will be normalized into relative dominance metrics where possible, such as:

- Win / success percentages
- Placement or ranking distributions
- Event-by-event performance consistency metrics

This normalization process allows disparate sports performances to be compared on a shared relative scale rather than through raw statistics alone.

Tools and Processing Workflow

- **Visual Studio Code (VS Code)** as the primary development environment for data cleaning, preprocessing, and visualization implementation
- **Relevant VS Code extensions** for CSV viewing/editing, code formatting, and project organization
- **Basic scripting and data manipulation techniques** to clean, merge, and transform datasets across multiple sources

Visualization of data

Proceed to the last page

Must have Features

A. Bar Chart Race

A bar chart race will be used to visualize cumulative points across all drivers throughout the 2023 season, directly supporting **Objective 1 (Verstappen's dominance across the season)**. This technique is effective because it highlights changes in ranking over time while preserving the magnitude of point differences, allowing users to clearly see when Verstappen separates from the field and how his lead expands dynamically.

B. Small Multiples (Line Charts)

Small multiples composed of line charts will be used to compare Verstappen and Sergio Pérez across key performance metrics, supporting **Objective 2 (driver vs.**

car influence). Line charts are well-suited for showing trends, consistency, and magnitude over time, which are critical for identifying performance differences between drivers. By separating metrics into small multiples, the design avoids clutter while enabling direct, side-by-side comparison across multiple aspects of performance.

C. Parallel Coordinates Plot

A parallel coordinates plot will be used to compare Verstappen's 2023 season with other dominant Formula 1 seasons, supporting **Objective 3 (historical comparison)**. This technique allows multiple performance metrics—such as win rate, podium percentage, and DNFs—to be visualized simultaneously, enabling users to evaluate how each season performs across several dimensions in a single view.

D. ECDF (Empirical Cumulative Distribution Function) Plot

An ECDF plot will be used to compare Verstappen's performance with elite athletes from other sports, supporting **Objective 4 (cross-sport comparison)**. This approach is effective because it represents the full distribution of performance rather than relying on summary statistics, allowing dominance to be interpreted through curve shape, where steeper and right-shifted curves indicate stronger and more consistent performance.

Optional Features

A. Tooltips with Detailed Performance Information

Interactive tooltips will provide additional contextual details when users hover over visual elements, such as race name, driver, finishing position, qualifying result, and points scored. This feature improves usability by allowing deeper exploration of the data without cluttering the primary visualizations.

B. Scrollytelling

The project may incorporate a scrollytelling interface to guide users through the analysis in a structured narrative sequence. As users scroll, visualizations can be introduced progressively with annotations and transitions, helping communicate the broader story of Verstappen's dominance in a more engaging and interpretable format.

C. Focus + Context

Users may be given the ability to filter specific drivers, seasons, or athletes across visualizations and highlight selected entities for focused comparison. This would improve interactivity and allow users to explore the data from perspectives beyond the default narrative.

Project Schedule

Week 1 (Apr 3 – Apr 10) – Proposal Phase

Deadline: Revised Proposal – Apr 10

- Finalize project topic (Verstappen 2023 analysis)
- Review and understand dataset (JDAustralia Kaggle dataset)
- Complete all proposal sections:
    - Objectives
    - Data
    - Data Processing
    - Visualization Design
- Develop initial sketches using the Five Design Sheet methodology
- Submit revised proposal

Week 2 (Apr 11 – Apr 17) – Initial Development

Deadline: Alpha Release – Apr 17

- Load dataset into VS Code
- Perform initial data verification and organization
- Compute key derived values (cumulative points, finishing positions, etc.)
- Create basic draft visualizations (line charts, bump charts)
- Begin implementing core components of the dashboard
- Submit Alpha Release with initial working visualizations

Week 3 (Apr 18 – Apr 24) – Core Feature Development

- Refine and finalize data structure

- Implement main visualizations:
    - Cumulative points progression
    - Race-by-race performance
- Begin implementing comparative driver analysis
- Improve layout and structure of visualization

Week 4 (Apr 25 – Apr 30) – Integration & Interaction

Deadline: Beta Release – Apr 30

- Add interactivity:
    - Driver selection
    - Highlighting and filtering
- Implement qualifying vs finishing analysis
- Integrate all visual components into one dashboard
- Improve design (colors, labels, readability)
- Submit Beta Release

Week 5 (May 1 – May 7) – Refinement & Polishing

- Debug and refine visualizations
- Improve usability and interaction flow
- Add optional features:
    - Tooltips
    - Annotations
    - Summary statistics
- Test for consistency and clarity

Week 6 (May 8 – May 14) – Presentation Preparation

Deadline: Project Presentation – May 11

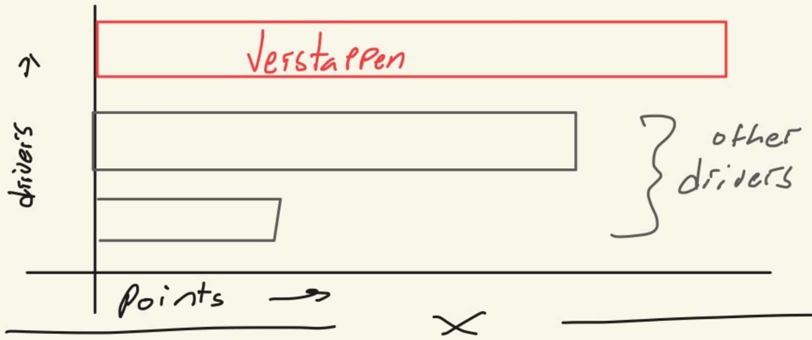
- Finalize visualization design
- Prepare presentation slides or demo
- Practice explaining insights and design decisions
- Deliver project presentation

Week 7 (May 15 – May 20) – Final Submission

Deadlines: Project Report Draft (May 17), Final Submission (May 20)

- Write and finalize project report
- Document methodology and insights
- Perform final polishing of visualization
- Submit final project

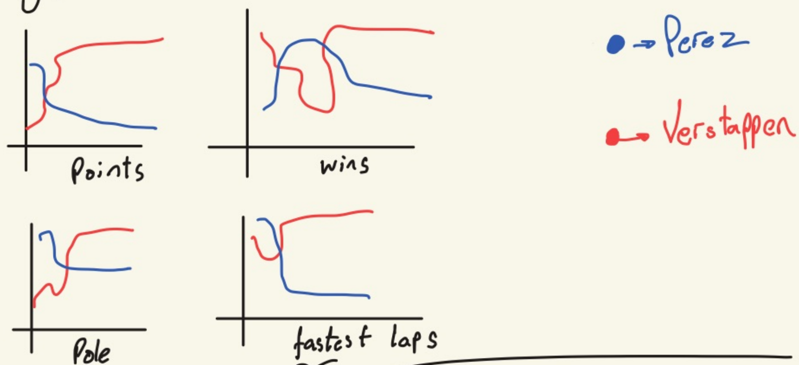
Prototype 1 → Perf Timeline



Bar chart Race :

→ Shows championship gap growth
→ best display of dominance

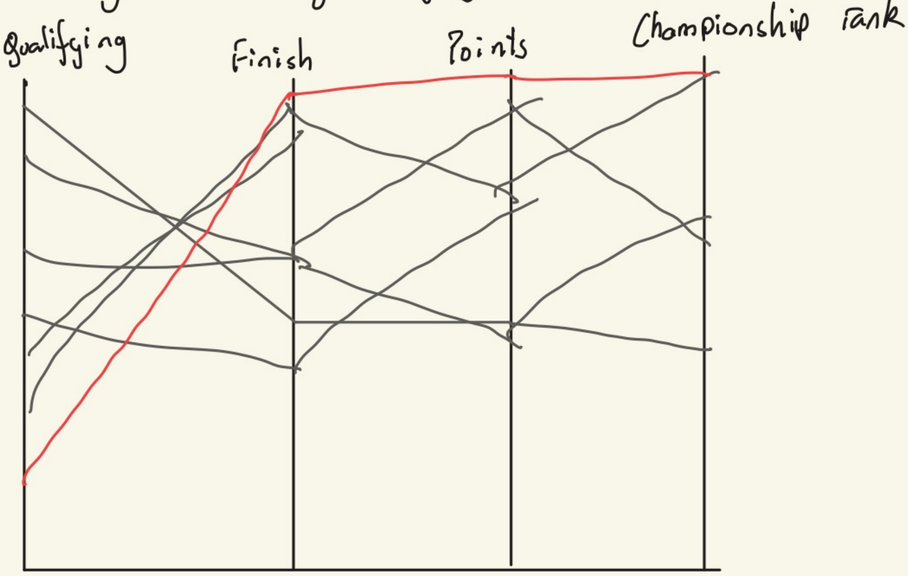
Prototype 2 → Championship Comparison



Small multiples :

→ Useful for multivariate comparison
→ Allows comparison for multiple factors.

Prototype 3 : Qualifying vs Race Perf



[I am still thinking the best options]

Parallel Coordinate Chart :

→ Captures multiple components
→ Easy to notice trends
→ Room for deeper analysis

Revised Section

Note: All revisions made relative to the original proposal have been highlighted in a sage color for clarity and ease of review.

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- "Don't Verstappen Me Now! Using Data Visualization to Benchmark Max Ver." *SportsChord*, 12 May 2023, www.sportschord.com/en-us/blogs/blog/dont-verstappen-me-now-using-data-visualization-to-benchmark-max-verstappen-against-the-f1-greats.
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Resources used for this

Only use of generative AI was to generate a rough project schedule and brush up words.

~ https://www.storytellingwithdata.com/blog/2020/1/6/swdchallenge-small-multiples

~ https://datajournalism.com/read/handbook/one/delivering-data/visualization-as-the-workhorse-of-data-journalism

~ https://blog.ouseful.info/2011/07/30/getting-my-eye-in-around-f1-quali-data-parallel-coordinate-plots-sort-of/

~ https://simulation.michelin.com/canopy/technical-articles/the-heart-of-f1

~ Reference for bump chart - https://public.tableau.com/app/profile/nastengraph/viz/MaxVerstappen10Formula1Wins/Bump

~ This reference for information and will perhaps be used for brushing up the dataset - https://www.formula1points.com/driver/Max%20Verstappen

~ Other websites looked at -

~ https://python.plainenglish.io/analyzing-the-performance-of-formula-1-drivers-in-the-2023-season-3f8d6f305cf5

~ https://www.sportschord.com/en-us/blogs/blog/dont-verstappen-me-now-using-data-visualization-to-benchmark-max-verstappen-against-the-f1-greats